Optoelectronics Lab – Experiment 3: Optical Signal Processing

Preparation: Task 1 and Questions 1–4

Name, Group: _____ Date: _____

Task 1

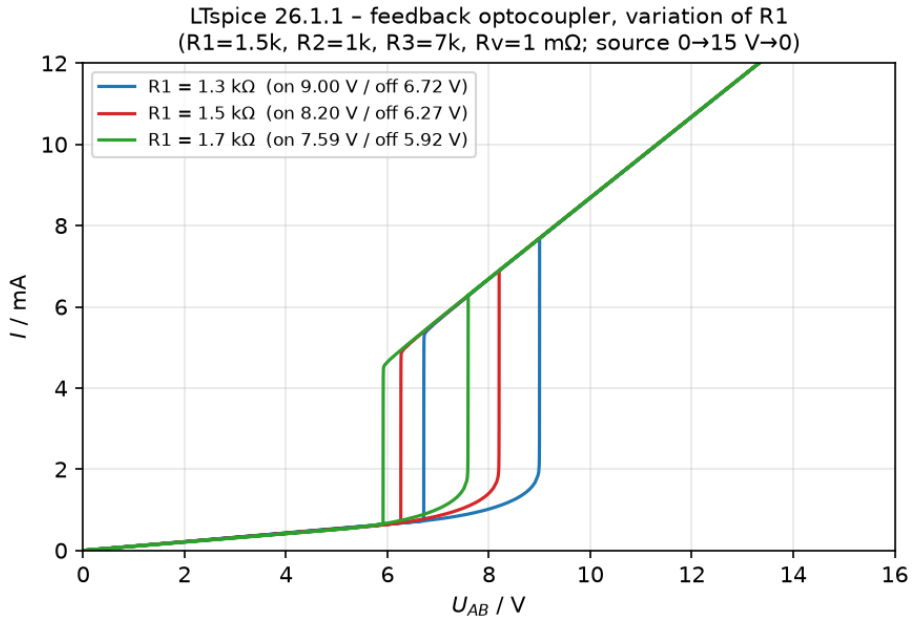
a) Simulation of the feedback optocoupler

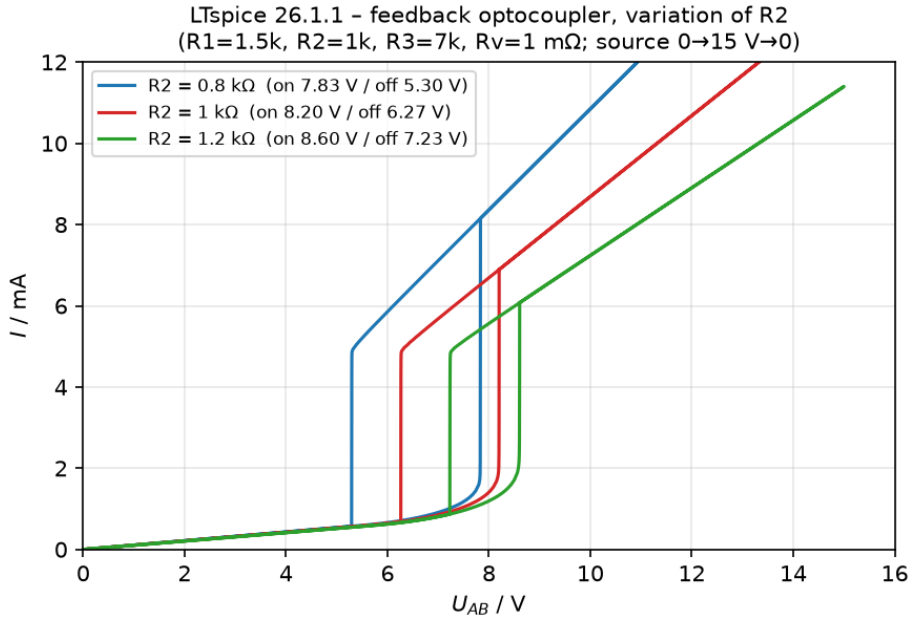
Simulation: LTspice 26.1.1 with the course files (Optokoppler_var_R1/R2/R3/Rv.asc; PC817 model with base drive I_B = 2e-3·I_{LED}, B_F = 1200 and B-E leakage I_{SE} = 3e-15 A). The source is swept 0 → 15 V → 0 (quasi-static, 1 s per ramp), the current I is plotted against U_{AB}. Basic values: R_1 = 1.5 kΩ, R_2 = 1 kΩ, R_3 = 7 kΩ, R_V = 1 mΩ.

Variation	Value	U_{on} / V (jump 1)	U_{off} / V (jump 2)	ΔU / V
R_1	1.3 kΩ	9.00	6.72	2.28
	1.5 kΩ	8.20	6.27	1.93
	1.7 kΩ	7.59	5.92	1.67
R_2	0.8 kΩ	7.83	5.30	2.53
	1.0 kΩ	8.20	6.27	1.93
	1.2 kΩ	8.60	7.23	1.37
R_3	6 kΩ	7.50	6.24	1.26
	7 kΩ	8.20	6.27	1.93
	9 kΩ	9.64	6.30	3.34
R_V (U_{source} at the jumps)	1 Ω	8.20	6.27	1.93
	100 Ω	8.40	6.75	1.65
	1 kΩ	–	–	no hysteresis

Table 1: Switching voltages of the feedback optocoupler (LTspice).

- **R_1:** the LED starts to conduct when $I \cdot R_1 \approx U_F$. On the OFF branch $I = U/(R_1+R_2+R_3)$, hence $U_{on} \approx U_F \cdot (R_1+R_2+R_3)/R_1$. Increasing R_1 lowers both switching voltages.
- **R_2:** in series in both states and dominant on the ON branch (slope ≈ $1/R_2$). Increasing R_2 raises both thresholds, flattens the ON branch and reduces the loop gain: the hysteresis narrows (2.53 V → 1.37 V).
- **R_3:** short-circuited by the phototransistor in the ON state, so it only sets the OFF-branch slope and U_{on}. U_{off} is practically unchanged, so the hysteresis widens with R_3 (1.26 V → 3.34 V).
- **R_V:** $U_{AB} = U_{source} - I \cdot R_V$ (load line). The jumps run along the load line with slope $-1/R_V$, while the fold points in U_{AB} stay at about 8.2 V / 6.3 V. The largest negative differential resistance of the curve is about 0.82 kΩ. For R_V > 0.82 kΩ (e.g. 1 kΩ) only one intersection exists, so the hysteresis disappears and the full S-shaped curve (branch 3 of Fig. 5) is traced.


LTspice 26.1.1 – feedback optocoupler, variation of R1
(R1=1.5k, R2=1k, R3=7k, Rv=1 mΩ; source 0→15 V→0)
R1 = 1.3 kΩ (on 9.00 V / off 6.72 V)
R1 = 1.5 kΩ (on 8.20 V / off 6.27 V)
R1 = 1.7 kΩ (on 7.59 V / off 5.92 V)
I / mA
U_{AB} / V
0
2
4
6
8
10
12
14
16


LTspice 26.1.1 – feedback optocoupler, variation of R2
(R1=1.5k, R2=1k, R3=7k, Rv=1 mΩ; source 0→15 V→0)
R2 = 0.8 kΩ (on 7.83 V / off 5.30 V)
R2 = 1 kΩ (on 8.20 V / off 6.27 V)
R2 = 1.2 kΩ (on 8.60 V / off 7.23 V)
I / mA
U_{AB} / V
0
2
4
6
8
10
12
14
16

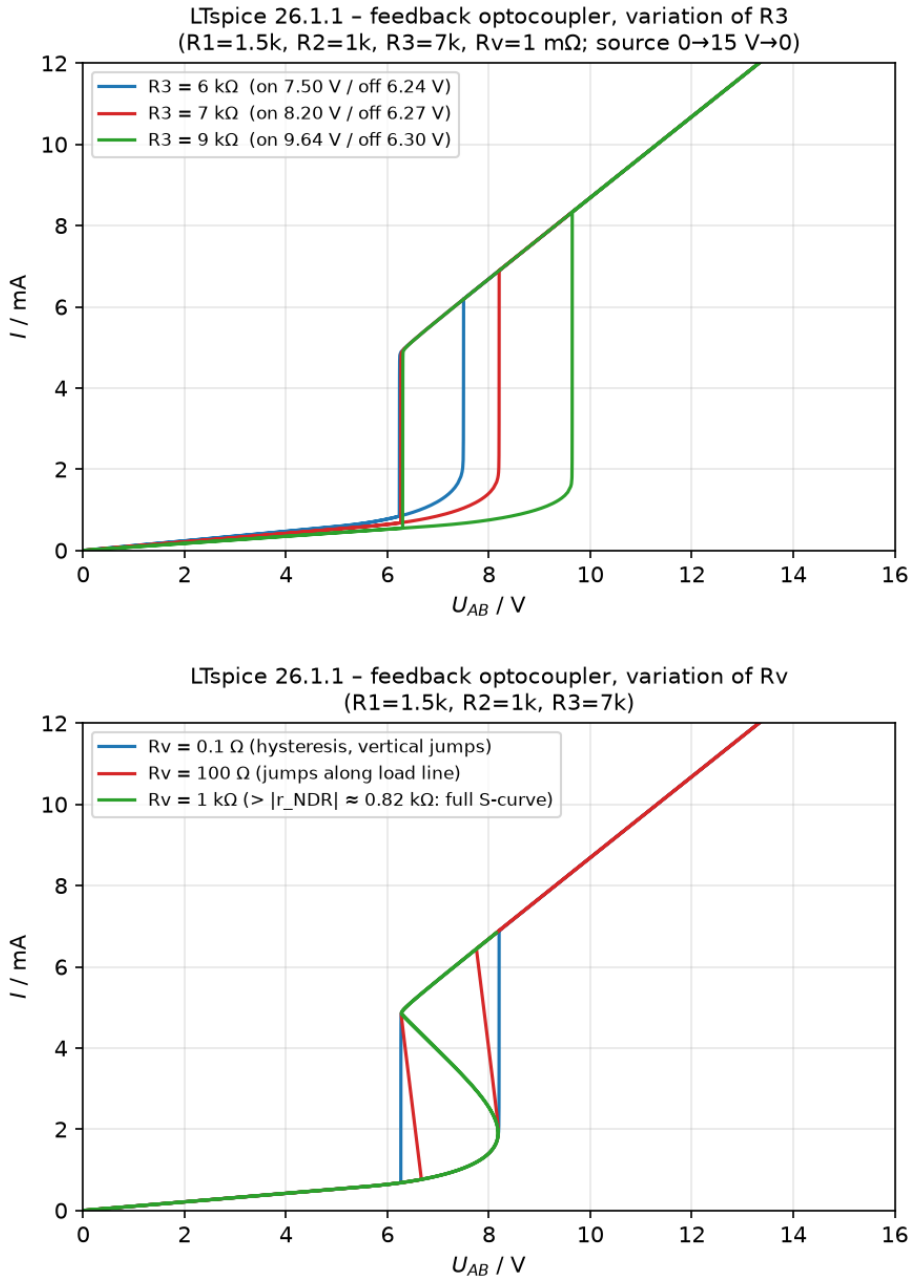
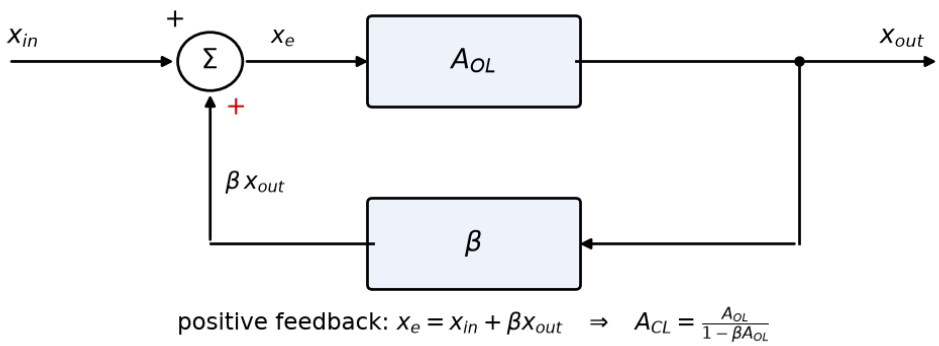


Fig. 1: I–U_{AB} characteristics simulated with LTspice.

b) Positive feedback – closed-loop gain



positive feedback: $x_e = x_{in} + \beta x_{out} \quad \Rightarrow \quad A_{CL} = \frac{A_{OL}}{1 - \beta A_{OL}}$

Fig. 2: Block diagram of positive feedback.

With $x_e = x_{in} + \beta \cdot x_{out}$ and $x_{out} = A_{OL} \cdot x_e$:

$$x_{out} = A_{OL} \cdot x_{in} + \beta \cdot A_{OL} \cdot x_{out} \quad \Rightarrow \quad \mathbf{A_{CL} = x_{out}/x_{in} = A_{OL} / (1 - \beta \cdot A_{OL})}$$

- $\beta \cdot A_{OL} < 1$: the gain is increased ($A_{CL} > A_{OL}$), the system is still stable.
- $\beta \cdot A_{OL} \to 1$: $A_{CL} \to \infty$; the system becomes unstable.
- $\beta \cdot A_{OL} > 1$: the output runs into saturation and stays there. This latching gives two stable states (bistability, hysteresis).

(For comparison, negative feedback gives $A_{CL} = A_{OL} / (1 + \beta \cdot A_{OL})$.) In the optocoupler, the loop LED → light → phototransistor → current through R_1 → LED is a positive feedback loop. Its loop gain is about $CTR \cdot R_3/(R_2+R_3)$. In the switching region the differential resistance is $dU/dI \approx R_2 + R_3 \cdot (1 - CTR)$, which becomes negative for $CTR \cdot R_3 > R_2 + R_3$ (loop gain > 1). Because the CTR of the phototransistor grows with the LED current, this condition is only met above a certain current, so the circuit switches.

c) NPN transistor at the operating point – load line

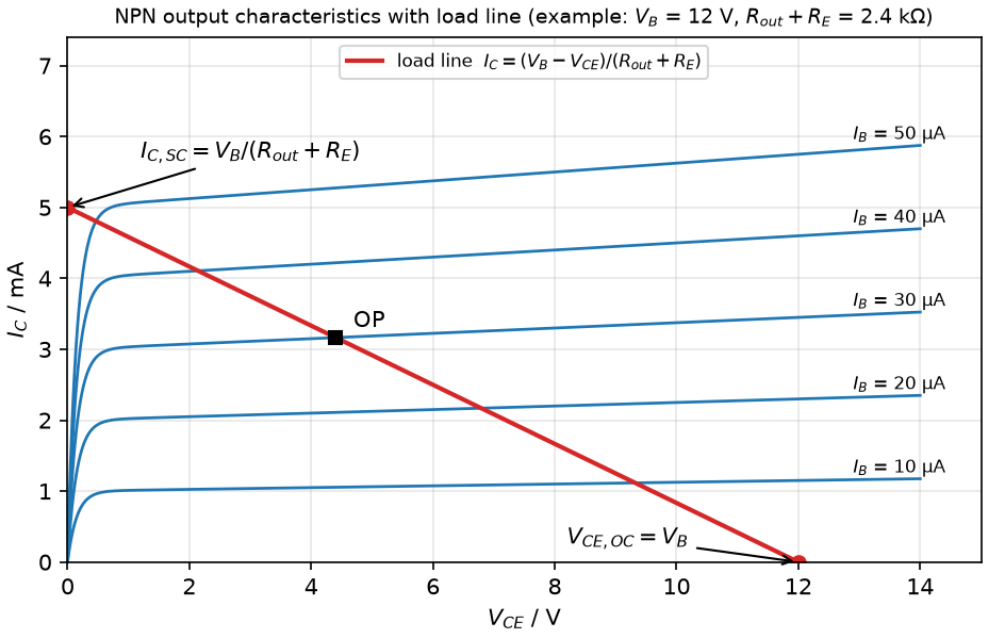


Fig. 3: Output characteristics $I_C(V_{CE})$ for different I_B with the load line of $R = R_{out} + R_E$.

Mesh equation (Fig. 6): $V_B = I_C \cdot R_{out} + V_{CE} + I_E \cdot R_E$. With $I_E = I_C + I_B \approx I_C$ ($B \gg 1$):

$$\mathbf{I_C = (V_B - V_{CE}) / (R_{out} + R_E)} \quad \text{(straight line with slope } -1/R\text{)}$$

- Open-circuit voltage ($I_C = 0$): $\mathbf{V_{CE,OC} = V_B}$
- Short-circuit current ($V_{CE} = 0$): $\mathbf{I_{C,SC} = V_B / (R_{out} + R_E)}$

The operating point is the intersection of the load line with the characteristic for the given I_B. The same construction is used for the load line of R_V in the I–U characteristic of the feedback optocoupler.

Question 1

a) Forward voltage: the voltage across the LED at which it carries a noticeable current. It is usually specified at the nominal current, e.g. 20 mA. Because the diode characteristic is exponential, this voltage stays almost constant. It is set mainly by the band gap: $U_F \approx E_g/e \approx h \cdot c/(e \cdot \lambda)$. Typical values: IR (GaAs, 940 nm) ≈ 1.2 V, red ≈ 1.8–2.0 V, green ≈ 2.1–3 V, blue/white ≈ 3–3.5 V. Temperature coefficient ≈ −2 mV/K.

b) Current-transfer ratio $CTR = I_C/I_F \cdot 100$ %. With a phototransistor the CTR is typically 20 % to several 100 % (PC817: 50–600 %, rank A: 80–160 %), because the photocurrent is amplified by the transistor current gain B. With a photodiode it is only about 0.1–1 %, because there is no internal gain. A phototransistor gives a high CTR but is slow (µs, Miller capacitance). A photodiode is fast (ns).

Question 2

a) Origin of the characteristic: at low voltage the LED and the phototransistor are off and $I = U/(R_1+R_2+R_3)$ (flat branch). When $I \cdot R_1$ reaches U_F, the LED emits and the phototransistor starts to conduct. It shunts R_3, so the current rises, the LED gets brighter and the phototransistor conducts more. This positive feedback makes the current jump to the ON branch. When the voltage is reduced, the phototransistor keeps R_3 short-circuited down to a lower voltage. The result is hysteresis, i.e. memory.

b) Ranges (1) and (2): below jump (1) the curve is set by $R_1+R_2+R_3$, and the jump is triggered by the LED threshold (U_F, R_1). Above jump (2), on the ON branch, the curve is set by R_2, the LED and $U_{CE,sat}$. The jump down happens when the phototransistor can no longer stay saturated ($CTR \cdot I_{LED} < I - U_{CE}/R_3$, i.e. the holding current is reached).

c) At the jump points: $dU/dI = 0$ (fold points) and the loop gain equals 1. The operating point merges with the unstable middle branch and disappears, so the circuit jumps along the load line to the other stable branch. In between lies a region of negative differential resistance, which is unstable under voltage control. Like a thyristor, the curve is S-shaped (current-controlled).

d) Influence of the resistors: see Task 1a. $R_1\uparrow$ lowers both thresholds; $R_2\uparrow$ raises them and flattens the ON branch; $R_3\uparrow$ raises U_{on} and widens the hysteresis; $R_V\uparrow$ tilts the load line, and for $R_V > |r_{NDR}| \approx 0.82\ k\Omega$ the S-curve becomes visible. Because U_{AB} is measured between A and B, R_V itself does not appear in the U_{AB} axis. It only decides which part of the curve can be reached.

Question 3

a) LED_2: in series with the whole circuit. It shifts the characteristic to the right by its forward voltage (≈ 1.7–2 V; LTspice: U_{on} 8.20 → 9.93 V, U_{off} 6.27 → 8.05 V) and adds a small series resistance. It also serves as the optical output: it lights up brightly only in the ON state.

b) Current and optical power: LED: $P_{opt} \approx \eta_{ext} \cdot (h \cdot \nu/e) \cdot I$, roughly linear (sublinear at high currents due to heating and droop). Photodiode: $I_{ph} = S \cdot P_{opt}$ with $S = \eta \cdot e/(h \cdot \nu)$ (≈ 0.5–0.6 A/W for Si near 850–900 nm), linear over many decades.

c) Illuminating PD_2: the photocurrent flows in parallel to R_3 into the loop, so the total current and $I \cdot R_1$ rise. The LED switches on and the positive feedback latches the circuit in the ON state (SET), which remains after the light is switched off. LTspice (with LED_2, $U_{source} = 9$ V, i.e. between the jump points 8.05 V and 9.93 V): an optical input equivalent to ≥ 0.3 mA input-LED current sets the memory (I: 0.96 → 5.9 mA); 0.2 mA is not sufficient.

d) LDR (light-dependent resistor, photoresistor, e.g. CdS): its resistance drops from MΩ in the dark to about 100 Ω under illumination. Connected in parallel to R_1, it reduces the voltage across the LED. The LED goes off, the phototransistor opens and the circuit returns to the OFF state (RESET). LTspice: reset for $R_{LDR} \lesssim 3\ k\Omega$ (5 kΩ is not sufficient).

e) Schmitt trigger: a comparator with positive feedback and two thresholds $U_{T+} > U_{T-}$ (hysteresis). It turns slow or noisy signals into clean digital edges and is used for debouncing, signal regeneration and relaxation oscillators. A plain comparator has only one threshold, so noise near the threshold causes multiple toggling (chatter). In the optical circuit, the memory and the Schmitt trigger differ only in the operating point. If the dark state is bistable, the circuit is a memory. If only OFF exists in the dark, it is a Schmitt trigger and needs a continuous input to stay ON. LTspice ($R_1 = 800\ \Omega$, $R_2 = 400\ \Omega$, $R_3 = 5\ k\Omega$, $U_{source} = 5$ V, $R_V = 0.1\ \Omega$): ON at $I_{in} \approx 1.24$ mA, OFF at ≈ 1.01 mA; the hysteresis narrows with R_V and disappears for $R_V \gtrsim 300\ \Omega$.

Question 4

a) A photodiode (light-sensitive element) in parallel to R_3, illuminated by the R and S signals.

b) $R_V = 0$: both elements see the same voltage and switch independently. For identical elements the result is one hysteresis with double current (LTspice: both ON at 9.94 V, OFF at 8.05 V). For different elements it is a twin hysteresis with two steps (R_1 = 1k, R_4 = 2k: 8.64 V and 12.54 V). Both elements can be ON at the same time, so the network is no flip-flop.

c) Load line R_V: it must be chosen so that (i) the state with **one** element ON is stable, i.e. $U_{AB} = U_{source} - I_1 \cdot R_V$ lies between U_{off} and U_{on}; and (ii) the state with **both** ON is unstable, i.e. $U_{source} - 2 \cdot I \cdot R_V < U_{off}$. Switching on one element then pulls U_{AB} so far down that the other one cannot stay ON. LTspice ($R_1 = R_4 = 1.5$ kΩ, $R_2 = 1$ kΩ, $R_3 = 7$ kΩ, LEDs in series): with $R_V = 450$ Ω the network works as an RS flip-flop for $U_{source} \approx 10.6$–11.6 V (light ≙ 1 mA input-LED current; 0.3 mA is too weak). With $R_V = 0.1$ Ω both LEDs stay ON after S and R, so the flip-flop fails.

R	S	$\bar{Q}$ (LED_1)	Q (LED_2)
0	0	$\bar{Q}_n$ (hold)	Q_n (hold)
1	0	1	0
0	1	0	1
1	1	forbidden	forbidden

Table 2: Truth table of the RS flip-flop (theory). For R = S = 1 the optical circuit drives both LEDs ON (Q = $\bar{Q}$ = 1); after both inputs are released the final state is a race (undefined).

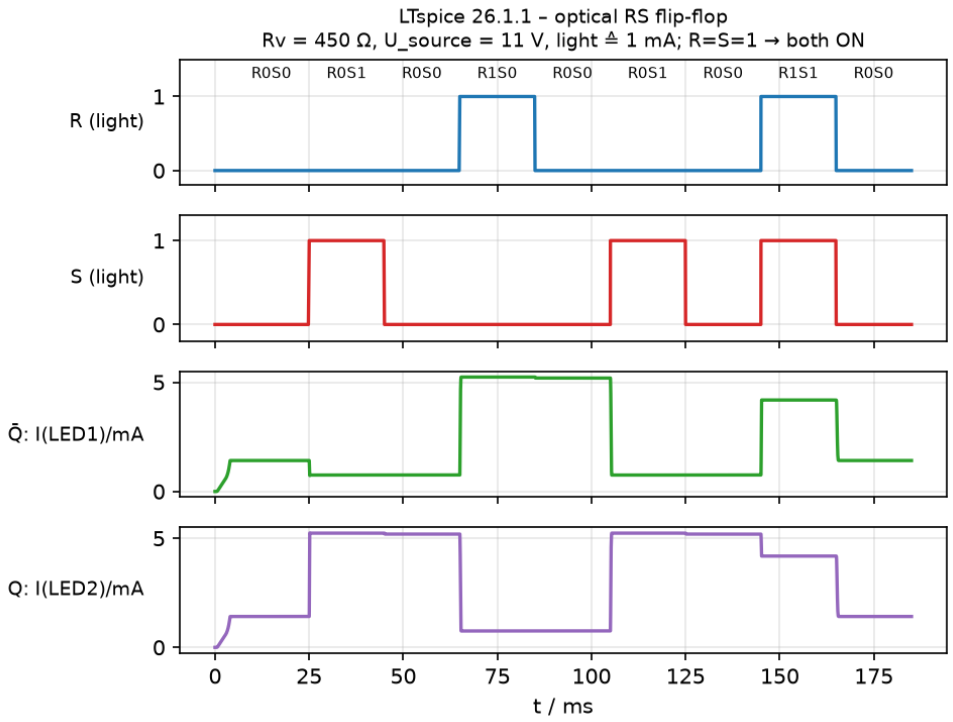


Fig. 4: Optical RS flip-flop simulated with LTspice ($R_V = 450$ Ω, $U_{source} = 11$ V).